

Which Head Takes Which Role? Initialization Leaves a Trace in Attention-Head Specialization

Anonymous ACL submission

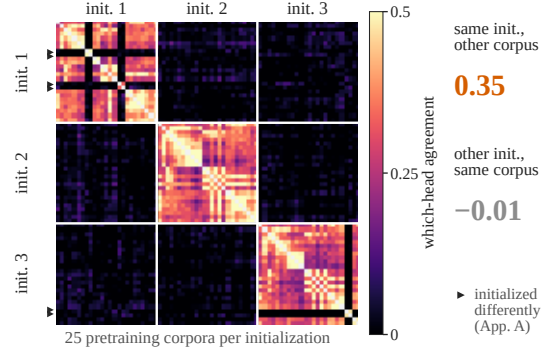
Abstract

Attention heads in a trained language model specialize into roles such as induction, previous-token and retrieval heads, but which head takes which role differs between training runs. How much of this difference comes from the random initialization and how much from the training data is unknown, because studies of training variability change only one of the two. We separate them with public model suites that train the same initializations on different corpora: DataDecide (three initializations, up to 25 corpora, 4M–1B parameters) and Pythia (one initialization, two versions of the Pile, 70M–12B). For nine head roles, the layer that hosts a role is shared by all models, while the head within that layer that takes the role is shared only by models with the same initialization, whatever their corpus. In controlled pretraining, this assignment is made during a short early stretch of training: perturbing the weights reassigns roles only within the first few percent of training, while switching the corpus rewrites a share of the assignment that shrinks the later the switch comes. How much of it two corpora share follows the similarity of their content words and the learning rate per batch token. The initialization decides where roles are placed, not what the model does: no initialization performs consistently better across corpora on 11 benchmarks, and removing the 1% of Flan data from pretraining changes how models use a counterfactual context after the template Question: while their head layout stays in place.

1 Introduction

Mechanistic interpretability describes a model through the roles of its components: an induction head completes a pattern seen earlier in the context (Olsson et al., 2022), a previous-token head attends to the preceding position, and retrieval heads copy a fact from the context into the prediction (Wu et al., 2025). When a model is trained again,

(a) Which head takes a role, 75 models at 1B



(b) Layer vs. head within the layer (9 roles)

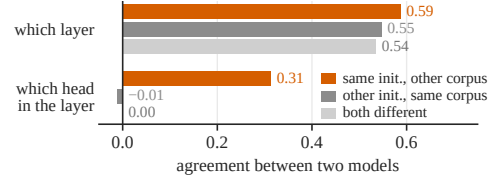


Figure 1: **Same initialization, same heads, on any corpus.** (a) Agreement on *which* head in a layer takes a role (within-layer rank correlation of head scores, mean over induction, previous-token and retrieval maps) between all 75 one-billion-parameter DataDecide models, three initializations $\times$ 25 corpora, ordered by initialization. Bright blocks: models that share an initialization agree on every corpus; models that share only a corpus do not. $\blacktriangleright$: six runs initialized differently from their label (§3), excluded from all means. (b) The two levels of a role map, mean over nine roles: the layer profile is shared by all pairs, the head within the layer only by pairs that share an initialization.

its algorithms tend to recur while the components that implement them change (Chughtai et al., 2023; Gurnee et al., 2024; Tigges et al., 2024; Bali et al., 2026). Two sources of this change are confounded. Studies of seed variability retrain on a fixed corpus (Sellam et al., 2022; van der Wal et al., 2025; Bali et al., 2026), and studies that relate mechanisms to data vary the data under one initialization or many unrelated ones (Chan et al., 2022; Singh

et al., 2024; Chen et al., 2026). Neither design can tell whether the head that takes a role is chosen by the initialization, by the data, or by neither.

We separate the two with a natural experiment that public pretraining suites already contain. DataDecide (Magnusson et al., 2025) trains three initializations on up to 25 corpora at each of 14 sizes, and Pythia (Biderman et al., 2023) trains one initialization on the Pile and on the deduplicated Pile at every size up to 12B parameters. Comparing models that share an initialization but not a corpus with models that share a corpus but not an initialization shows how much each factor contributes. For nine head roles we ask two questions: which layer hosts the role, and which head within that layer takes it. Heads within a layer are interchangeable, so independently initialized models agree on the second question only by chance, which gives every comparison a baseline of zero (§2).

The two questions have different answers (Figure 1). All models agree on the layer. Only models that share an initialization agree on the head within the layer, and they do so on any corpus: under one initialization, the same head of layer 7 is the strongest induction head on 14 of 25 corpora. The agreement holds for all nine roles, at all 14 DataDecide sizes and in Pythia up to 12B, and it is specific enough to tell which initialization a model came from. Comparing layouts without labels, we found six DataDecide checkpoints that, as their released weights confirm, were initialized differently from their label (§3).

We then ask when the assignment is made and what decides how much of it carries over to another corpus. In controlled pretraining, switching the corpus or perturbing the weights during the first 1% of training reassigns the roles. A few percent later, a model returns to its assignment even after noise as large as its weights, and a switch of corpus rewrites only part of it, less the later it comes. The window closes after a fixed amount of learning rather than a fixed number of steps, and it does not depend on the optimizer state (§4). How much of the assignment two corpora share follows the similarity of their content words, and how strongly the initialization decides it at all follows the learning rate per batch token, which also explains why larger DataDecide models agree more (§5).

Finally, we ask what the initialization does not decide (§6). How strong a circuit becomes, when it forms and how a model scores on 11 benchmarks follow the corpus, and no initialization does consis-

tently better across corpora. A case study on one data component makes the division concrete. Removing Flan, about 1% of Dolma 1.7, changes how models use a counterfactual context introduced by Flan’s template `Question:`, while the head layout of these models stays as close to its seed-matched siblings as under any other change of corpus. Almost a third of the attention to the context that Flan adds after `Question:` falls on the model’s ten strongest retrieval heads, 4% of its heads, whose placement the initialization fixes: the data teaches the habit, and the habit acts in part through heads that the initialization placed.

Our contributions are:

- A seed $\times$ corpus design built from public suites at 21 sizes from 4M to 12B parameters, with a measurement that separates which layer from which head hosts a role and a permutation-symmetry baseline (§2).
- Evidence that a shared initialization fixes which head takes each role across corpora, at every size and for every role we measure, together with an audit of the suites’ initializations (§3).
- Controlled experiments that locate when the assignment is fixed and identify corpus content and the stochastic-gradient noise scale as what determines how much of it is kept (§4, §5).
- A delineation of what the initialization does not determine, with a case study of a template habit that 1% of the pretraining data installs on top of an unchanged head layout, and its consequences for knowledge-conflict evaluation (§6).

2 Setup and Measurement

Initializations crossed with corpora. DataDecide pretrains OLMo-style models at 14 sizes (4M–1B parameters) on 25 corpus recipes—C4, DCLM and its quality-filtered variants, Dolma 1.6/1.7 and their ablations, Falcon RefinedWeb variants, FineWeb-Edu/-Pro and mixtures—each with three seeds. Within a size, a seed’s initial weights are shared across recipes, so each initialization has been trained on up to 25 corpora. Pythia trains a standard and a deduplicated model from identical initial weights at every size; we use 70M–12B, with the independently initialized PolyPythias (van der Wal et al., 2025) as a reference. We check every run’s initialization against its released weights (Appendix A). Model pairs fall into three types: *same initialization*, *other corpus* (seed-matched pairs);

other initialization, same corpus; and both different.

Head-role maps. For each model we compute a [layer $\times$ head] map for each of nine roles from fixed probe material (Appendix B): induction, previous-token, retrieval (attention from the answer position to the context entity in knowledge-conflict prompts), attention sink (Xiao et al., 2024), duplicate-token, current-token, two-back and delimiter heads, and each head’s OV copying score, which is computed from the weights alone (Elhage et al., 2021).

Two levels of agreement. A role map says which layer hosts a role and which head within the layer takes it. We compare two models on the first with the rank correlation of their per-layer profiles and on the second with the *which-head agreement*: the Spearman correlation of head scores within each layer, averaged over layers.

The chance baseline. Permuting the heads of a layer, together with their query, key, value and output weights, leaves the network’s function unchanged; the initialization distribution and gradient-based training respect this symmetry. Two independently initialized models therefore agree on which head takes a role only by chance, and their expected which-head agreement is exactly zero, whatever their corpora (Lemma 1, Appendix C). Agreement above zero between models trained on different corpora thus measures what their shared initialization carries through training.

Behaviour. We use DataDecide’s 11 benchmark evaluations, circuit strength and developmental timing, and *context trust*: on knowledge-conflict prompts that state a counterfactual fact, the log-probability margin of the counterfactual over the memorized answer, on facts the model knows (Li et al., 2025; Du et al., 2024).

3 Initialization Fixes Which Head Takes a Role

Same initialization, same heads. In the 1B crossing, models with the same initialization agree on which head within a layer takes each role, while models with different initializations agree at chance, even on the same corpus (Table 1a, Figure 1). This holds for all nine roles (0.18–0.43 against -0.02 to 0.00), including the copying score computed from the weights alone, with five ini-

(a) DataDecide 1B, 3 seeds $\times$ 25 corpora (69 models)

Role	same init.	other init.	gap, 95% CI
Two-back	0.43	0.00	[0.40, 0.46]
Previous-token	0.42	-0.01	[0.40, 0.47]
Duplicate-token	0.36	-0.02	[0.36, 0.41]
Induction	0.33	-0.02	[0.32, 0.37]
Current-token	0.32	-0.02	[0.31, 0.36]
Retrieval	0.31	0.00	[0.28, 0.35]
Delimiter	0.22	-0.01	[0.20, 0.25]
Attention sink	0.18	-0.01	[0.16, 0.23]
OV copying (weights)	0.25	-0.01	[0.25, 0.29]

(b) Pythia, one initialization: Pile vs. deduplicated Pile

Size	induction	prev.-token	retrieval	null
70M	–	0.60	0.58	0.26
160M	0.49	0.58	0.53	0.14
410M	0.47	0.55	0.45	0.09
1B	0.56	0.72	0.58	0.16
1.4B	0.48	0.54	0.50	0.09
6.9B	0.53	0.60	0.49	0.06
12B	0.38	0.51	0.38	0.04
12B, at 2.1%	0.55	0.60	0.45	0.04

Table 1: **Models with the same initialization agree on which head takes each role.** Which-head agreement (within-layer Spearman of head scores). (a) All nine roles at 1B for models with the same initialization trained on different corpora and with different initializations trained on the same corpus; the gap’s interval is a recipe bootstrap. (b) Pythia’s two models from one initialization; null: 95th percentile after permuting heads within layers; no induction head at 70M.

tializations at 1B, and on corpus pairs that share no data source. When every head’s within-layer rank is decomposed into initialization, corpus and residual components, the initialization is significant for 66–79% of heads and the corpus for 4–7%, close to the 5% false-positive rate; the median corpus component is zero. The agreement holds at all 14 DataDecide sizes and grows with size (Figure 2a; Spearman with log parameters 0.81–0.92). In Pythia, the two models from one initialization agree at every size from 70M to 12B, far above a permutation null (Table 1b), while independently initialized PolyPythias agree at chance.

The layer is shared by all models. The layer profile of each role agrees at 0.4–0.7 for every pair type (Figure 1b), and all 75 1B models implement in-context copying with the same two-step algorithm, a previous-token head composing with an induction head. Where in depth a role lives and how it is computed are common to all models; the initialization decides which of the interchangeable heads in a layer carries it.

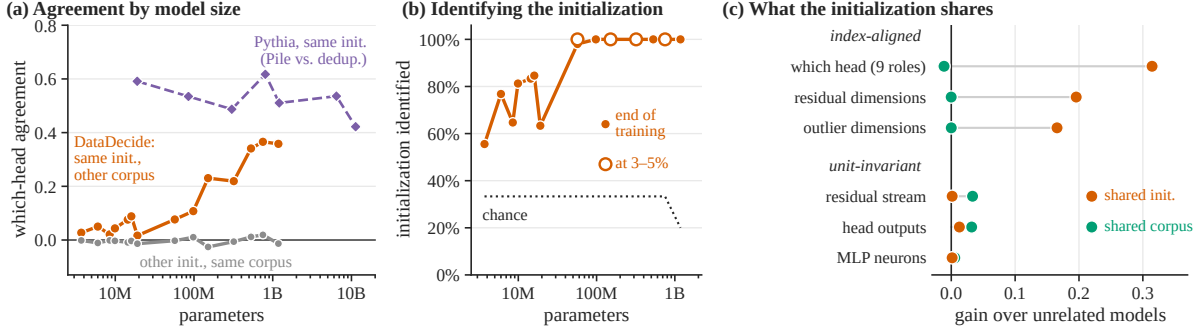


Figure 2: **The trace across sizes, as an identifier, and in other quantities.** (a) Which-head agreement (mean over induction, previous-token and retrieval maps) for DataDecide pairs with the same initialization and different corpora and with different initializations and the same corpus, at 14 sizes, and for Pythia’s two models from one initialization, up to 12B. (b) Identifying a model’s initialization from its head layout, leaving its corpus out, at the end of training and at 3–5% of training. (c) Similarity gained over unrelated models by sharing the initialization or the corpus: the initialization is shared in index-aligned quantities, the corpus in unit-invariant ones.

The layout is specific to the initialization. We hold out a model, compare its head layout with the models trained from each initialization on the *other* corpora, and assign it to the closest one. From 60M parameters on, this identifies the initialization with 98–100% accuracy (chance: 33%, or 20% with five seeds), including all 69 verified 1B models, and already at 3–5% of training (Figure 2b, Table 6); in Pythia, the true sibling ranks first among ten candidates at all three sizes with independently initialized alternatives. Applied without labels to the 75 1B models, the layout singles out six runs that agree with none of their nominal siblings but pairwise with each other (Figure 1a, ►). Their released weights confirm that these runs were not initialized from their seed’s weights (correlation 0.000) and that each pair shares an initialization of its own (Table 5).

What carries the trace. The weights themselves keep a small trace: same-initialization models trained on different corpora have weights that correlate at 0.01–0.04, against 0.000 for other pairs, which identifies the initialization at every size, as does the output-bias fingerprint SeedPrints (Tong et al., 2026) from 60M on (Table 7). Beyond this, the models have drifted apart. Final weights correlate with their initial values at 0.03–0.09, same-initialization models on different corpora are not linearly connected (loss barrier 7.5 nats, against 7.8 for unrelated models; Frankle et al., 2020), and under unit-invariant measures, CKA (Kornblith et al., 2019) and best-match similarity, their representations are no closer than those of unrelated models, while models that share a corpus are clos-

est (Figure 2c). What the initialization shares are index-aligned quantities: which head hosts a role, and which residual dimensions carry most variance (0.20 vs. 0.00) and the massive outlier activations (0.17 vs. 0.00); MLP neurons do not keep their identity (0.008). The head layout is the functional form of this trace: it says which component does what.

4 When the Assignment Is Fixed

Intervening during training. We pretrain small Llama-style models on C4 and branch them at step k of 10,000, either switching the corpus to source code or adding Gaussian noise to every weight (Figure 3a, Table 2). Up to $k = 100$ (1% of training), both interventions reassign the roles: after a switch to code, agreement with the uninterrupted run is 0.09, and noise as large as the weights leaves 0.03 at step 0 and 0.36 at step 100. From $k = 250$ (2.5%) on, the model returns to its assignment after the same noise (0.85 at 2.5%, 0.95–0.98 from 5%, the level of an exact rerun). A switch to code still rewrites part of the layout, a smaller part the later it comes: agreement is 0.60 at 2.5%, 0.71 at 10% and 0.95 at 80%. A 12-layer model behaves alike: a switch to code at 1% keeps 0.40 of the layout and at 10% 0.84, while training on code from the start keeps none.

What closes the window. The early steps are also the warm-up of the learning rate (300 steps), and the branches inherit the optimizer’s state, so we repeat the noise branches with a 1,000-step warm-up and, separately, with a freshly initialized

Step	Share	300-step warm-up				1,000-step warm-up		Optimizer reset
		Switch to code	Noise 100%	Noise 10%	Code, 12 layers	Switch to code	Noise 100%	Noise 100%
0	0%	−0.03	0.03	0.74	−0.03	—	—	—
100	1%	0.09	0.36	0.94	0.40	—	0.16	—
250	2.5%	0.60	0.85	0.97	—	0.41	0.70	0.81
500	5%	0.72	0.96	0.99	—	—	0.87	0.91
1000	10%	0.71	0.95	0.99	0.84	0.69	0.87	0.97
2000	20%	0.81	0.98	1.00	—	—	0.94	—
4000	40%	0.86	0.96	1.00	—	—	—	—
8000	80%	0.95	0.98	1.00	—	—	—	—
exact rerun					0.97			

Table 2: **Interventions during controlled pretraining.** Which-head agreement (mean of induction and previous-token maps) between the final layout of a run intervened on at a given step and the uninterrupted run, for the default 300-step learning-rate warm-up, a 1,000-step warm-up, and branches that start with a freshly initialized optimizer. Step 0 for the code switch: the same initialization trained on code from scratch; noise is Gaussian, scaled to each tensor’s standard deviation.

optimizer (Appendix G). With the longer warm-up the window closes later in steps (0.70 at 2.5%, 0.87 from 5%), and the two schedules fall on one curve when agreement is plotted against the learning applied before the intervention, the cumulative learning rate (Figure 6). Resetting the optimizer changes little (0.81, 0.91 and 0.97 at 2.5, 5 and 10%). The assignment is thus fixed by a small, early amount of learning and is held by the weights themselves.

Public suites. The public models also settle early (Figure 3b). The same-initialization agreement of the DataDecide 1B models reaches its final level by 3.6% of training, Pythia’s previous-token heads stabilize by about 2%, and Pythia-12B’s two models already share their layout at 2.1% (Table 1b). Before training, no statistic of a head’s initial weights or initial gradient predicts which head will take a role, so the assignment is not readable at step 0; it is set by the first steps of training on top of the initialization.

5 What Decides How Much Is Kept

The agreement between same-initialization models varies: across corpus pairs, across sizes and across training setups. Two quantities account for most of this variation.

Corpus similarity. Across DataDecide’s corpus pairs, the Jensen–Shannon distance between two corpora’s unigram distributions predicts how closely an initialization’s layout recurs on both. The relation holds at every size from 60M on and

strengthens with size (Spearman -0.20 to -0.33 at 60M, -0.65 to -0.83 from 300M on, -0.76 to -0.80 across the 300 corpus pairs at 1B; Mantel $p < 0.001$ from 90M; Table 6), on pairs that share no data source (-0.36 to -0.82), and after partialling out source overlap. Bigram distance predicts less (-0.55 to -0.62 at 1B). Splitting the vocabulary into function words (Yang et al., 2026) and content words shows which part of the distribution matters (Figure 4). Content-word distance predicts agreement at every size (ρ from -0.23 to -0.69), also when function-word distance is controlled (partial $\rho -0.25$ to -0.78), whereas function-word distance adds nothing once content words are controlled (partial $\rho \geq 0.09$); the English web corpora in DataDecide use function words alike. Controlled pretraining extends the relation to more distant corpora: C4 and scientific papers share the layout (0.19 ± 0.06 over nine initializations), pairs involving source code share none (≈ 0), and Pythia’s two versions of the Pile, nearly identical in content, share it most.

Gradient noise. Which head takes a role is settled when the symmetry between heads breaks early in training, by the initialization together with the gradient noise of those steps. The noise of stochastic gradients scales with the learning rate per example in a batch (Smith and Le, 2018; Jas- trzębski et al., 2018), and lowering it lets the initialization decide more. At fixed model size, runs that differ only in the order of their batches agree 0.43 with 16-sequence batches and 0.90 with 512-sequence batches, and same-initialization agree-

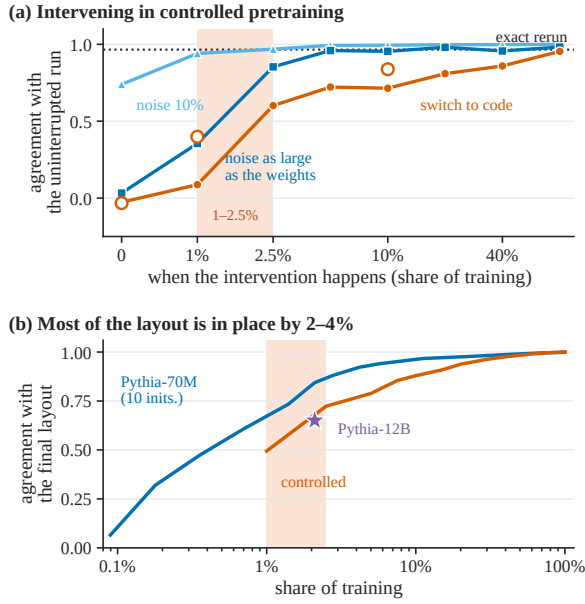


Figure 3: **The assignment is fixed early.** (a) Controlled pretraining (4 layers $\times$ 8 heads, 10k steps): agreement of the final layout with the uninterrupted run after switching the corpus to code or adding Gaussian noise to every weight at a given step; hollow circles: a 12-layer model. (b) Agreement of the layout with its own final layout during training, in public suites and controlled runs; star: Pythia-12B at 2.1%. Shaded: 1–2.5% of training.

ment between C4 and scientific papers rises from 0.01 to 0.22 (Appendix H); tripling the learning rate acts like quartering the batch. This accounts for the growth with size in DataDecide: its recipes lower the learning rate per batch token 140-fold from 4M to 1B, whereas a 30 $\times$ larger controlled model at a fixed rate keeps no more of its initialization, and Pythia, trained with large batches at every size, keeps it at 70M.

6 What the Initialization Does Not Decide

Strength, timing and performance follow the corpus. How strongly a circuit is expressed follows the corpus (significant corpus effect on 4 of 6 strength measures, initialization component $\approx$ 0), and so does when it forms: in controlled pre-training with six initializations and three corpora, the corpus explains 81% of the variance in the onset of induction heads and the initialization none ($p = 0.0001$ vs. 0.47). Across 11 benchmarks $\times$ 14 sizes, the initialization has a significant main effect in 3.9% (accuracy) and 7.8% (per-byte likelihood) of the 154 cells, about the 5% expected by chance, and the corpus in 75% and 98% (median

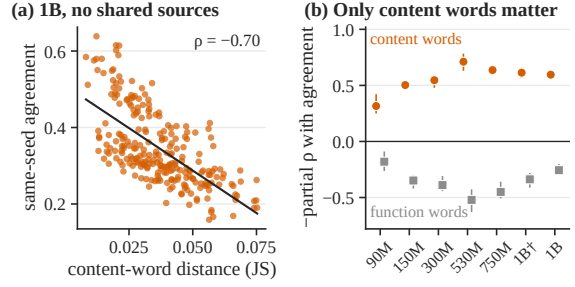


Figure 4: **Similar content words, similar heads.** (a) 1B pairs of DataDecide corpora that share no data source: Jensen–Shannon distance between their content-word distributions vs. the which-head agreement of same-initialization models trained on them. (b) Negative partial Spearman correlation between distance and agreement for content words (controlling for function words) and for function words (controlling for content words), by size; bars span the three roles; †: 1B with five seeds at 10% of training.

Property	Follows	Key statistic
Which head takes a role	initialization	0.18–0.43 vs. 0.00
Residual coordinates	initialization	0.20 vs. 0.00
Layer that hosts a role	all models	0.4–0.7, all pairs
Copying algorithm	all models	75 / 75 models
Circuit strength	corpus	4 / 6 measures
Onset of induction heads	corpus	81% vs. 0%
Representations	corpus	CKA, best match
11 benchmarks, 14 sizes	corpus	init. sig. 4–8%
Knowledge conflicts	corpus	init. sig. 0 / 12
Question: habit	corpus	+2.4 nats

Table 3: **Which properties follow the initialization and which the corpus.** Key statistic: same- vs. different-initialization agreement; variance share of corpus vs. initialization; or share of cells with a significant initialization effect (5% is the false-positive rate).

variance components: initialization 0.000, corpus 0.49–0.83; Table 9); behaviour under knowledge conflict shows no initialization effect in any of 12 conditions. The variance that the initialization does contribute is interaction with the corpus: an initialization that does well on one corpus is no more likely to do well on another (Table 3).

6.1 Case Study: A Data Component and the Question: Template

The previous results concern where roles are placed and show that performance does not follow the placement. This case study asks the converse question for one data component that clearly changes behaviour: does it move the head layout, and where is the behaviour it teaches expressed? We compare

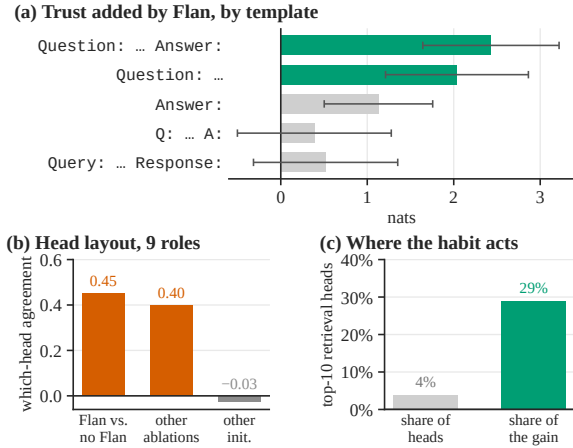


Figure 5: **Flan changes behaviour, not the head layout** (1B). (a) Flan minus no-Flan models: added trust in a counterfactual context relative to a declarative prompt, by template ($\pm$ SE over initializations). (b) Which-head agreement (mean over nine roles) of same-initialization pairs with and without Flan, of same-initialization pairs of other Dolma 1.7 ablations, and of different-initialization pairs. (c) Share of the attention to the context entity that Flan adds after Question: (relative to Q:) falling on each model’s ten strongest retrieval heads, against their share of all heads.

seed-matched models trained with and without the component. Dolma 1.7 contains about 1–2% of Flan instruction data (Longpre et al., 2023; Soldaini et al., 2024), and DataDecide trains a Dolma 1.7 recipe with Flan removed from the same initializations. We compare the two on knowledge-conflict prompts from ParaConflict (Li et al., 2025): one sentence asserts a counterfactual fact (“The capital of France is Rome.”) and is followed by a declarative continuation or by a question in Flan’s format (Question: What is the capital of France? Answer:).

Flan teaches trust after Question:. Removing Flan lowers the 1B models’ trust in the counterfactual context by 1.3–3.5 nats across four kinds of context when the prompt is a question, and leaves declarative prompts and the models’ knowledge of the facts unchanged. The share of questions on which a model ignores a one-sentence context and answers from memory rises from 1.9% to 14.3%. The effect is tied to Flan’s own wording (Figure 5a, Table 4): Question: alone carries most of it, whereas the equivalent Q: ... A: and Query: ... Response: leave trust unchanged, although Q: is three times as frequent as Question: in Dolma 1.7; a question about a different entity

Ending after the context	count in Dolma 1.7	Flan effect	SE
Question: q Answer:	11.3M / 9.6M	+2.43	0.79
Question: q	11.3M	+2.04	0.83
Answer:	9.6M	+1.13	0.63
Q: q A:	38.4M / 6.9M	+0.39	0.89
Query: q Response:	0.15M / 0.75M	+0.52	0.83
Question: q' Answer:	as first row	+3.06	1.42
declarative (no question)	–	+0.20	0.50

Table 4: **The habit is keyed to Flan’s literal template.** Flan minus no-Flan models (1B, three initializations each): added trust in the counterfactual context relative to a declarative ending, in nats. Example: *The capital of France is Rome.* Question: *What is the capital of France?* Answer: *The capital of France is ...*; q' asks about another entity. Marker counts: occurrences in the pretraining corpus. The last row is the raw trust difference for the declarative ending.

triggers it as strongly. The habit appears at six of seven sizes from 60M to 1B, is present by 3.6% of pretraining, replicates on PopQA counterfactuals (Mallen et al., 2023) and NQ-Swap (Longpre et al., 2021), and appears in OLMo 2 (Team OLMo et al., 2024) in each of three mid-training runs on a mixture containing FLAN (Appendix J).

The layout does not move. Seed-matched models with and without Flan agree on which head takes each role at 0.45 (mean over nine roles), as closely as seed-matched pairs of other Dolma 1.7 ablations (0.40), while different initializations agree at -0.03 (Figure 5b). Removing Flan changes the behaviour and leaves the assignment of roles where the initialization put it.

Where the habit acts. To locate the habit, we measure each head’s attention from the last prompt token to the counterfactual entity, the read-out of the retrieval role, and take the difference between Question: and Q: prompts. Models trained with Flan attend more to the context after Question: (summed over heads, $+0.54$), models without Flan less (-0.21). Of this added attention, 29% falls on the model’s ten strongest retrieval heads, which make up 4% of its heads (Figure 5c), and models with the same initialization agree on where it falls (0.11, against -0.03 for different initializations). The habit is learned from the data and expressed in part through the retrieval heads that the initialization placed.

Consequences for evaluation. Context-faithfulness and knowledge-conflict benchmarks

commonly pose questions in exactly this format, and every public base model we tested trusts a counterfactual context more after Question: than in a declarative prompt, by 1.0–4.6 nats, most of all the model pretrained with Flan (Table 11). A score on such benchmarks therefore mixes faithfulness with a habit that a small slice of pretraining data installs and a change of template removes, which gives prompt-template sensitivity (Sclar et al., 2024; Hua et al., 2025; Liu and Chu, 2026) a concrete source in the pretraining corpus.

7 Related Work

Seeds and training variability. Seed variation is a known source of irreproducibility in fine-tuning and pretraining (Reimers and Gurevych, 2017; Dodge et al., 2020; McCoy et al., 2020; Sellam et al., 2022), modelled as noise (Benito-Santos et al., 2025) or studied as convergence across training (Fehlauer et al., 2025; van der Wal et al., 2025); SeedPrints (Tong et al., 2026) identifies a seed from output biases. Varying only the seed, these studies cannot separate the initialization from the data; crossing the two shows that the initialization fixes the placement of head roles but not what the model does.

Universality and stability of circuits. Algorithms recur across models while components vary (Chughtai et al., 2023; Gurnee et al., 2024; Tigges et al., 2024), attention heads differ across retraining (Bali et al., 2026), specialize along developmental trajectories (Wang et al., 2025), and can be read from their parameters (Elhelo and Geva, 2025). Our results locate the variation: the layer and the algorithm are shared, and the head within a layer follows the initialization.

Data and the formation of mechanisms. Data properties decide whether in-context learning emerges (Chan et al., 2022; Singh et al., 2024), and individual training examples shape when a head forms (Chen et al., 2026). We find that the corpus sets whether, when and how strongly a mechanism forms, while the initialization sets where. Early training is known to be decisive for what a network can still learn (Achille et al., 2019) and for which basin it ends in (Frankle et al., 2020); we show the same for which head takes which role.

Knowledge conflicts and prompt format. Whether a model follows a counterfactual context or its memory depends on the context’s form and

the model’s confidence (Longpre et al., 2021; Xie et al., 2024; Du et al., 2024), is mediated by competing mechanisms (Ortu et al., 2024), and shifts with the prompt’s wording (Zhou et al., 2023; Sclar et al., 2024; Hua et al., 2025; Liu and Chu, 2026); synthetic pretraining shows which data properties make models use both sources (Kim et al., 2026). We trace one such format effect to a 1% slice of natural pretraining data.

8 Discussion and Conclusion

Interpretability. A circuit found in one model generalizes to other models in its algorithm and its layers; its head labels transfer only to models with the same initialization trained on similar corpora. Ablating the ten most important heads of a same-initialization sibling trained on another corpus does 19.5% of the damage of ablating the model’s own; heads found in a different-initialization model trained on the *same* corpus do 2.0%. Comparisons of circuits across training data should therefore match initializations, and seed-matched siblings in public suites provide such comparisons: their heads correspond, so differences between their circuits can be attributed to the data.

Seed-crossed suites. Suites that cross initializations with data make these comparisons possible and are only as informative as their seed labels; comparing siblings is a cheap check of the labels, and releasing a hash of each run’s starting weights would make it unnecessary.

Training and evaluation. Lower gradient noise, through larger batches or smaller learning rates, makes head assignments reproducible across corpora, and changing the corpus early in training draws a new one. On the evaluation side, the absence of a transferable initialization effect means that selecting seeds does not carry over between recipes and that mixed-effects analyses should model initialization $\times$ corpus interaction.

Conclusion. Models trained from one initialization on different corpora share which head takes each role. The assignment is fixed by a small amount of early learning, outlasts the drift of the weights away from their initial values, and recurs to the extent that corpora share content words and training is low in gradient noise. What the heads do, from their strength and timing to benchmark performance and habits such as trust after Question:, follows the corpus.

Limitations

We study decoder-only transformers pretrained on English corpora, up to 12B parameters, and define head roles through attention patterns and OV circuits. Multilingual seed-crossed suites would test the corpus-similarity relation where function words differ across corpora, and other architectures can be added as such suites appear. Our interventions during training use models of up to 12 layers; intervening on larger models in their first percent of training is the natural next step. The template habit is one data component; why models key on Question: rather than an equivalent marker, and which other instruction-data families install habits of their own, are open. Which property of the initialization decides which head will take a role is likewise open.

Ethical Considerations

We analyse publicly released models and datasets under their licences and train small models on public corpora; no human subjects are involved. We report our audit of public checkpoints to support their correct use.

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A Auditing the Public Suites

DataDecide 1B recipe	seed	step 0 vs. seed’s step 0	step 2500 vs. seed’s step 0	step 2500 vs. partner
DCLM QC 7% FW2	1	0.00	0.00	0.11
DCLM QC 7% FW3	1	0.00	0.00	0.11
Dolma 1.7 no math/code	1	0.00	0.00	0.08
Dolma 1.7 no Reddit	1	0.00	0.00	0.08
Falcon+CC QC orig. 10%	3	0.00	0.00	0.09
Falcon+CC QC Tulu 10%	3	0.00	0.00	0.09
genuine siblings (C4 vs. Dolma 1.7, same seed), step 2500				0.07–0.08

Table 5: **Six 1B runs were not initialized from their seed’s weights.** Correlation of one attention weight matrix (block 5) between checkpoints. The other 69 runs share their seed’s step-0 weights exactly.

Step-0 identity. Pythia’s standard and deduplicated models have tensor-identical step-0 weights at 70M, 160M, 410M, 1.4B and 2.8B, rounding-level identical weights at 1B (correlation 0.99993) and identical files at 6.9B and 12B. For DataDecide we compare one weight matrix of every released step-0 checkpoint with the step 0 of the same seed’s C4 run, reading the tensor directly from the hosted files.

All sizes. Below 1B, we use for each size the recipes whose three seeds share their published step-0 weights (by file hash) and the checkpoint listed in Table 6: the final checkpoint up to 90M, and at larger sizes a checkpoint available for all selected runs (43–98% of training; 11% for the five-seed 1B set). For every one of these 686 runs we compare the earliest training checkpoint (step 1,250; 2,500 at 1B) with the step-0 weights of every seed. Each correlates with its own seed’s step 0 (0.05–0.41, rising with size) and with no other seed’s (≤ 0.009), except one 1B run at 11% of

training (FineWeb-Pro, seed 3), which matches no seed and is excluded.

Training start. A published step-0 checkpoint need not be the state a run started from, so we compare the earliest training checkpoint with every candidate initialization. The DataDecide 1B runs of C4, Dolma 1.7 and DCLM correlate at step 2,500 only with their own seed’s step-0 weights (0.23 vs. 0.000). PolyPythias’ “weight-seed” variants at 160M, in contrast, were trained from the standard initialization: their step-1 weights equal the standard model’s (correlation 1.0) and are unrelated to their own published step 0; we label them accordingly.

Six 1B runs with an unlisted initialization. Six 1B DataDecide runs disagree with all their nominal siblings in Figure 1a: the DCLM-QC 7% FineWeb-2/3 recipes and the Dolma 1.7 no-math-code / no-Reddit ablations of seed 1, and the Falcon-and-CC QC original / Tulu recipes of seed 3 (Table 5). Their published step-0 weights differ from their seed’s, their step-2,500 weights correlate with no seed’s step 0, and each pair’s step-2,500 weights correlate with each other as strongly as those of two genuine siblings: each pair was trained from an initialization of its own. All other 69 runs share their seed’s step-0 weights exactly, so the layout flagged exactly the six runs that differ. We exclude them from every same- and different-initialization comparison at 1B.

A near-duplicate Pythia pair. The released final checkpoint of pythia-2.8b-deduped is nearly identical to that of pythia-2.8b (attention-weight correlation 0.993, embedding correlation 0.985), whereas models trained from one initialization on different data retain a correlation of only 0.03–0.09 with it; we exclude the 2.8B pair.

B Head Roles

Induction: attention from the second copy of a random 128-token block to the token after its first occurrence (200 blocks). **Previous-token:** attention to position $t - 1$ on 50 natural texts (split-half reliability 0.99–1.00). **Attention sink:** attention to position 0 on the same texts. **Retrieval:** attention from the last prompt token to the counterfactual entity in the knowledge-conflict prompts. **Duplicate-token, current-token, two-back and delimiter:** attention to an earlier copy of the current token, to itself, to $t - 2$ and to punctuation, on natural

text. **OV copying:** the share of positive eigenvalue mass of the head’s OV circuit composed with the embedding and unembedding (Elhage et al., 2021), computed from weights alone. An induction map is used only when the model has an induction head (maximum score > 0.3).

C Head Permutation Symmetry and the Chance Baseline

Lemma 1. *Let a layer have H heads and let π permute them together with their query, key, value and output blocks. Assume (i) the initialization distribution is invariant under π ; (ii) training is π -equivariant, i.e. the update applied to $\pi\theta$ is π applied to the update of θ (true for gradient descent, Adam/AdamW, weight decay and global-norm clipping, as the loss is π -invariant); and (iii) the data and their order are independent of the initialization. Let $w(\theta)$ be the head with the largest score for a role, computed from the network’s internals, so that $w(\pi\theta) = \pi(w(\theta))$. Then for every corpus D , $P(w(\theta_T) = h \mid D) = 1/H$, and two independently initialized models trained on any corpora D, D' agree on the winning head with probability $1/H$; the expected within-layer rank correlation of their head scores is 0.*

Proof sketch. By (i)–(ii), θ_T and $\pi\theta_T$ have the same distribution given D , so $w(\theta_T)$ is uniform over heads. Independent initializations make w and w' independent, so they agree with probability $\sum_h (1/H)^2 = 1/H$; averaging over permutations sends any rank correlation to zero in expectation. $\square$

The lemma fixes the baseline for pairs with independent initializations; it makes no prediction for pairs that share one, whose agreement is the quantity we measure. Different-initialization pairs agree at -0.02 to 0.00 for all nine roles (Table 1a).

D Results at Every Size

Table 6 lists, for every DataDecide size, the recipes and seeds used, same- and different-initialization agreement, initialization identification at the end of training and in the earliest checkpoints, the corpus-distance relation and the recipe’s learning rate per batch token. Agreement is weak up to 20M parameters, where most models have not yet formed an induction head, and strong from 150M on.

Parameters	Checkpoint (share)	Recipes	Seeds	Same init., other corpus	Other init., same corpus	Init. ID (%)	Init. ID at 3–9% (%)	ρ (corpus distance, agreement)	LR per batch token (10^{-9})
3.7M	5,725 (100%)	18	3	0.03	0.00	56	–	–	213.6
6.0M	9,182 (100%)	23	3	0.05	–0.01	77	–	–	183.1
8.5M	13,039 (100%)	17	3	0.02	0.00	65	–	–	167.8
9.9M	15,117 (100%)	16	3	0.04	0.00	81	72	–	152.6
14M	21,953 (100%)	18	3	0.08	–0.01	83	–	–	140.4
16M	24,432 (100%)	13	3	0.09	0.00	85	–	–	135.8
19M	14,584 (100%)	20	3	0.02	–0.01	63	61	–	64.1
57M	29,042 (100%)	17	3	0.08	0.00	98	100	–0.20 to –0.33	29.5
98M	29,901 (100%)	20	3	0.11	0.01	100	–	–0.43 to –0.50	15.0
151M	37,500 (98%)	15	3	0.23	–0.03	100	100	–0.47 to –0.64	10.7
320M	45,000 (98%)	15	3	0.22	–0.01	100	100	–0.69 to –0.78	5.0
530M	51,250 (89%)	10	3	0.34	0.01	100	–	–0.65 to –0.71	3.1
750M	27,500 (43%)	10	3	0.37	0.02	100	100	–0.66 to –0.77	2.1
1.2B [†]	7,500 (11%)	10	5	0.36	–0.01	100	–	–0.75 to –0.83	1.5
1.2B	69,369 (100%)	25	3	0.35	–0.01	100	–	–0.76 to –0.80	1.5

Table 6: **DataDecide at every size.** The checkpoint used (DataDecide revision step<N>–seed–<seed>; share of the default seed’s training), the recipes whose three seeds share their step-0 weights, which-head agreement (mean over induction, previous-token and retrieval maps; induction only where models have induction heads), leave-one-corpus-out identification of the initialization from the head layout (chance 33%; 20% with five seeds), the Spearman correlation between unigram corpus distance and same-initialization agreement over corpus pairs (range over roles), and the recipe’s learning rate per batch token. †: five seeds at 11% of training; last row: the verified three-seed crossing at the end of training. Every run starts from its labelled initialization (Appendix A).

E Identifying the Initialization

Table 7 compares identification from the head layout with two baselines on the same leave-one-corpus-out task. Final weights identify the initialization at every size: same-initialization pairs keep a weight correlation of 0.01–0.04, other pairs 0.000. SeedPrints (Tong et al., 2026), a rank test on final hidden states for random token sequences, identifies it from 60M on; its statistic is also raised for pairs that share only a corpus, which the head layout and the weights are not. On the six 1B runs with an unlisted initialization, all three methods agree: each run is closest to its partner run and to none of its nominal siblings.

F Controlled Pretraining

Llama-style models (RMSNorm, SwiGLU, RoPE, untied embeddings, OLMo tokenizer): 4 layers $\times$ 8 heads, $d = 256$; and 12 layers $\times$ 12 heads, $d = 768$. AdamW (learning rate 10^{-3} , $\beta = (0.9, 0.95)$, weight decay 0.1), 300 warm-up steps, cosine decay to 10%, batches of 64×512 tokens, 10,000 steps. Corpora: the first 400M tokens of single DataDecide sources for web text (C4), code (StarCoder), scientific papers (peS2o) and books (Gutenberg). Branches continue from the parent’s saved model and optimizer state at step $k \in \{100, 250, 500, 1000, 2000, 4000, 8000\}$, either switching the corpus or adding $\epsilon \cdot \text{std}(W) \cdot$

$\mathcal{N}(0, 1)$ to every tensor (Table 2). At initialization the basin is finite: perturbations up to 1% of the weight scale leave the layout unchanged (0.91–0.99), 10% partly reassign it (0.57), and 100% reassign it fully (0.05–0.18).

G Warm-up and Optimizer State

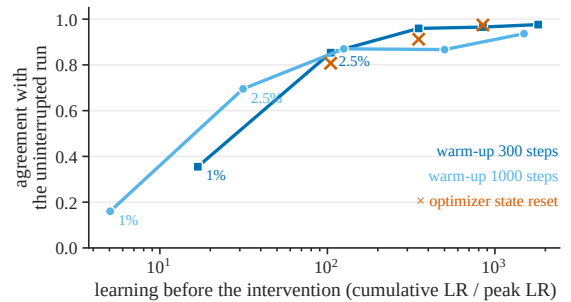


Figure 6: **The window closes after a fixed amount of learning.** Agreement of the final layout with the uninterrupted run after noise as large as the weights at 1, 2.5, 5, 10 and 20% of training, against the cumulative learning rate before the intervention (in units of the peak rate), for warm-ups of 300 and 1,000 steps; crosses: branches with a freshly initialized optimizer (300-step warm-up). Means over two initializations and the induction and previous-token maps.

We repeat the noise branches of Appendix F with two parents whose learning rate warms up over 1,000 instead of 300 steps, branching at $k \in$

Size	Identified (%)			Weight correlation same init.	SeedPrints z	
	head layout	weights	SeedPrints		same init.	same corpus
3.7M	56	98	94	0.022	9.5	1.0
6.0M	77	100	91	0.016	8.7	0.7
8.5M	65	100	94	0.013	8.7	1.6
9.9M	81	100	88	0.011	8.4	1.5
14M	83	100	94	0.009	7.8	3.6
16M	85	100	82	0.009	7.1	3.4
19M	63	100	97	0.016	8.1	2.3
57M	98	100	100	0.010	11.9	5.1
98M	100	100	100	0.012	13.3	3.9
151M	100	100	100	0.013	24.0	5.1
320M	100	100	100	0.016	30.1	5.9
530M	100	100	100	0.019	46.7	6.1
750M	100	100	100	0.022	46.4	4.1
1.2B [†]	100	100	100	0.043	54.7	1.7
1.2B	100	100	100	0.016	63.8	9.8

Table 7: **Identifying the initialization three ways.** Leave-one-corpus-out accuracy from the head layout, from the correlation of final weights (a fixed sample of 2M attention and MLP weights) and from SeedPrints (Tong et al., 2026); mean weight correlation of same-initialization pairs, and mean SeedPrints statistic of same-initialization and of same-corpus, different-initialization pairs. [†]: five seeds at 11% of training.

{100, 250, 500, 1000, 2000}, and with branches that discard the parent’s AdamW state at $k \in \{250, 500, 1000\}$. A parent continued from its own step-2,000 state reproduces its final layout (0.995–1.00), so the comparisons are not limited by run-to-run noise. With the longer warm-up, a switch to code keeps 0.41 of the layout at 2.5% and 0.69 at 10% (0.60 and 0.71 with the default warm-up). Changing only the warm-up, with the same initialization and data order, changes part of the final layout (agreement 0.21–0.59 for induction and 0.62–0.67 for previous-token maps), another sign that the first steps of training help decide which head takes a role.

H Gradient Noise

Batch	LR	LR per batch token (10^{-6})	other batch order	C4 vs. papers
16	$1 \cdot 10^{-3}$	0.12	0.43	0.01
64	$3 \cdot 10^{-3}$	0.09	0.57	0.07
64	$1 \cdot 10^{-3}$	0.03	0.61	0.21
64	$3 \cdot 10^{-4}$	0.01	0.82	0.19
512	$1 \cdot 10^{-3}$	0.00	0.90	0.22

Table 8: **Gradient noise.** Same-initialization which-head agreement of previous-token maps at step 2,000 between runs that differ only in batch order, and between runs on C4 and on scientific papers.

At fixed size we vary the batch (16, 64, 512 sequences) and the learning rate ($3 \cdot 10^{-3}$, 10^{-3} , $3 \cdot 10^{-4}$) and read the layout at step 2,000, after the

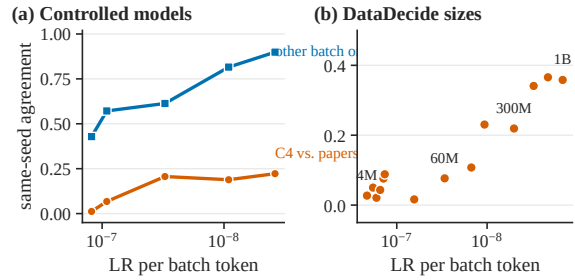


Figure 7: **Less gradient noise, more agreement.** (a) Controlled models at step 2,000: which-head agreement between runs that differ only in batch order, and between C4 and scientific-paper runs from one initialization, against the learning rate per batch token. (b) DataDecide: same-initialization agreement against the recipe’s learning rate per batch token.

assignment is fixed (Table 8). Order-only agreement is ordered by the learning rate per batch token, and tripling the learning rate acts like quartering the batch.

I Initialization Effects on Benchmarks

Table 9 breaks the benchmark analysis of §6 down by task. The corpus is a significant factor on almost every task and size; the initialization reaches significance in 0–2 of 14 sizes per task, as expected by chance, and its median variance share is zero on every task.

Benchmark	Accuracy				Correct-answer likelihood per byte			
	init. sig.	corpus sig.	init. share	corpus share	init. sig.	corpus sig.	init. share	corpus share
ARC-Easy	0/14	14/14	0.00	0.95	0/14	14/14	0.00	0.91
ARC-Challenge	1/14	10/14	0.00	0.49	0/14	14/14	0.00	0.92
BoolQ	0/14	7/14	0.00	0.25	2/14	13/14	0.00	0.55
CSQA	0/14	12/14	0.00	0.55	1/14	14/14	0.01	0.56
HellaSwag	2/14	13/14	0.00	0.60	2/14	14/14	0.00	0.96
MMLU	1/14	13/14	0.00	0.87	1/14	14/14	0.00	0.95
OpenBookQA	1/14	10/14	0.00	0.40	1/14	14/14	0.00	0.85
PIQA	0/14	14/14	0.00	0.56	1/14	14/14	0.00	0.76
Social IQa	0/14	10/14	0.00	0.30	2/14	13/14	0.00	0.48
WinoGrande	1/14	3/14	0.00	0.07	1/14	13/14	0.00	0.79
OLMES average	0/14	10/14	0.00	0.46	1/14	14/14	0.00	0.78

Table 9: **Initialization effects on benchmarks.** For each benchmark, the number of the 14 model sizes at which the initialization or the corpus has a significant main effect ($p < 0.05$, two-way ANOVA over recipes and seeds) and the median unbiased variance share of each factor across sizes.

J The Template Habit in Detail

Prompts. Context S = the first sentence of a Para-Conflict substitution conflict; declarative: S + answer stem; QA: S + Question: q Answer: + stem; variants replace the markers (Table 4).

Items. Facts known to every compared model (the memorized answer is more likely than the counterfactual without context): 1,680 ParaConflict items over six relations, 1,299 PopQA items, and 3,533 (DataDecide) and 3,750 (OLMo 2) NQ-Swap items. Standard errors use the seed-to-seed variance pooled over four 1B recipes with three seeds each.

Across sizes, training and datasets. The habit is present at six of seven sizes from 60M to 1B (Figure 8a, Table 10), appears by 3.6% of pretraining and grows to +2.4 nats (Figure 8b), and replicates on PopQA counterfactuals (+1.91) and on NQ-Swap, where on natural reading-comprehension passages it is strongest after Question: and extends to Q: (Figure 8d). In OLMo 2, mid-training on a mixture containing FLAN raises it from ≈ 0 to +1.4–2.3 nats in each of three runs, while trust under declarative prompts falls (Figure 8c); the mixture contains other data as well, so this replication shows the habit in a second training stack rather than isolating FLAN. Continued pretraining of a Flan-free 60M model on 90M Flan tokens does not install it ($+0.04 \pm 0.04$).

Public models. Table 11 gives the trust of public base models with and without the question format.

K Artifacts, Compute and Software

Artifacts. We use the DataDecide models (Apache 2.0) and data recipes (ODC-BY), Pythia and PolyPythias (Apache 2.0), OLMo 2 (Apache 2.0), ParaConflict (Apache 2.0), NQ-Swap (MIT) and PopQA as released by their authors, all for research use consistent with their intended use; the corpora are English.

Compute. All experiments ran on NVIDIA A100 80GB GPUs, about 150 GPU-hours in total, including exploratory runs; most of it went to the controlled pretraining runs, while measuring the public checkpoints takes a few minutes per model.

Software. PyTorch 2.8, Hugging Face Transformers 4.57 and Datasets 4.8, NumPy 1.26 and SciPy 1.17; models are loaded in bfloat16 with eager attention to read attention patterns. Benchmark scores are DataDecide’s released OLMES evaluations (Magnusson et al., 2025); marker counts in Dolma 1.7 come from the infini-gram index.

L Statistics

Recipe-bootstrap 95% intervals for same- minus different-initialization agreement; unbiased two-way variance components from mean squares with F tests; permutation tests with recipe labels permuted; Mantel tests for pair-level regressions (Mantel, 1967); partial Spearman correlations by rank regression.

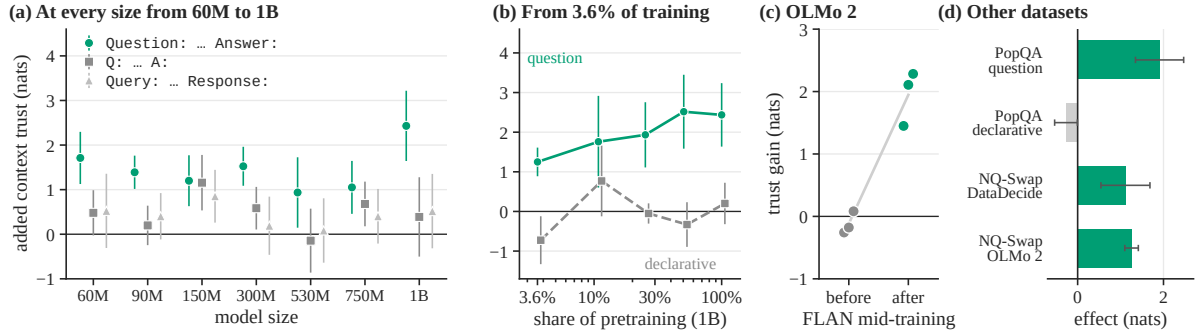


Figure 8: **The Question: habit across sizes, training and datasets.** Added trust in a counterfactual context (log-probability margin of the counterfactual over the memorized answer, relative to a declarative prompt; nats, $\pm$ SE). (a) Flan minus no-Flan DataDecide models at every size, for the Flan template and two equivalent templates. (b) During pretraining of the 1B models: present at the first checkpoint (3.6%); declarative prompts unaffected. (c) OLMo 2 1B before and after mid-training on a mixture with 16.6% FLAN (three mid-training runs). (d) PopQA counterfactuals and NQ-Swap (Question: vs. Q:).

Size	Question:qAnswer:	Question:q	Answer:	Q:qA:	Query:qResp.:	other entity
60M	+1.71 (0.58)	+0.73 (0.35)	+1.74 (0.59)	+0.48 (0.51)	+0.52 (0.83)	+2.14 (0.51)
90M	+1.39 (0.37)	+1.23 (0.55)	+1.51 (0.17)	+0.20 (0.44)	+0.40 (0.52)	+1.89 (0.47)
150M	+1.20 (0.57)	+1.09 (0.43)	+1.45 (0.70)	+1.16 (0.62)	+0.85 (0.59)	+1.48 (0.41)
300M	+1.52 (0.44)	+1.60 (0.60)	+1.54 (0.50)	+0.58 (0.48)	+0.19 (0.65)	+1.37 (0.41)
530M	+0.94 (0.79)	+0.96 (0.79)	+0.33 (0.70)	-0.14 (0.72)	+0.08 (0.72)	+1.51 (1.14)
750M	+1.05 (0.59)	+1.72 (0.35)	+1.35 (1.40)	+0.68 (0.50)	+0.40 (0.61)	+2.17 (0.50)
1B	+2.43 (0.79)	+2.04 (0.83)	+1.13 (0.63)	+0.39 (0.89)	+0.52 (0.83)	+3.06 (1.42)

Table 10: **The Question: habit at every size.** Flan minus no-Flan DataDecide models: added trust in the counterfactual context relative to a declarative ending, in nats (SE). The Flan template carries the effect at every size; equivalent templates do not.

Model	Instruction data in pretraining	declarative	question	gain
OLMo-1B (July 2024)	Dolma 1.7, with Flan	4.33	8.92	+4.60
Qwen2.5-1.5B	synthetic data from instruct models	-0.47	2.70	+3.18
Pythia-1B	none (the Pile)	5.47	6.89	+1.41
Pythia-1.4B	none (the Pile)	5.41	6.78	+1.38
SmolLM2-1.7B	not documented	2.89	3.93	+1.04
Gemma-2-2B	not documented	3.48	4.51	+1.03
OLMo-2-1B, before mid-training	none			-0.26 to +0.08
OLMo-2-1B, after mid-training	16.6% FLAN			+1.45 to +2.28

Table 11: **The habit in public base models.** Trust in a one-sentence counterfactual context (nats) with a declarative ending and with a Question:/Answer: ending, and the gain from the question format. The model pretrained with Flan trusts a question most.